A working model of behavior leakage: a constant write, gated by context similarity

June 2026

1 Setup

We study *leakage*: a behavior trained into a model under one context (a system prompt, persona, or ICL prefix) also appearing under other, untrained contexts. The measured quantity is the trained−base log-probability of the behavior token(s) at the end of a bystander context's own on-policy response.

Represent objects as residual-stream activation vectors at a fixed layer:

- a **context** c by $v_c \in \mathbb{R}^d$, read at the last prompt token (mean over a probe set of user questions);
- a **behavior** b by $v_b \in \mathbb{R}^d$, read at the position where the behavior is emitted (for the marker, the end-of-response slot).

Write f for the model's (nonlinear, unknown) map from a context to the behavior representation it produces there, so $v_b = f(c)$ before training. Nothing below assumes f is linear.

2 What the literature establishes

Five premises, each independently supported.

L1 (weights read and write linear features).
The residual stream is a linear communication channel: every weight matrix interacts with it by *reading* a set of input directions and *writing* a set of output directions [1]. A rank-one matrix $uw^\top$ is the atomic read/write unit: it reads feature w (a dot product) and writes feature u (a scaled addition). LoRA makes this factorization literal: in $\Delta W = BA$, the rows of A are read directions ("if" filters) and the columns of B are write directions; a rank-1 adapter projects the activation onto its A vector to produce a scalar gate on its B vector, which acts as a steering vector on the residual stream [2].

L2 (behaviors are linear write features).
Adding a single direction to the residual stream elicits a behavior (steering vectors [3, 4]); persona traits are readable as single directions at the last prompt token, with projection predicting trait expression before generation [5]; a single "toxic persona" direction most strongly predicts and controls emergent misalignment [6]; independently trained misaligned fine-tunes converge to one shared linear misalignment direction, and EM can be induced by as few as nine rank-1 adapters [7].

L3 (narrow fine-tunes add a constant write, gated by pre-existing reads).
Narrow LoRA fine-tunes "essentially add a constant steering vector": the added vectors have

pairwise cosines near $|1|$ across prompts and tokens — including on entirely unrelated out-of-distribution text — and extracting that direction and adding it as an explicit unconditional steering vector reproduces the fine-tune's behavior [8]. Even a trigger-gated (conditional) backdoor is implemented by a constant vector added at a *fixed* position, with the conditionality supplied by pre-existing attention circuitry (§5). Narrow fine-tuning also leaves a context-independent activation bias on unrelated text, whose rank-1 projection is causally load-bearing [9]. Conditionality, in this picture, is *magnitude along a fixed write direction*, not a new gated circuit.

L4 (edits are rank-one key→value associations).
Linear associative memories store input–output pairs as outer products, and interference between stored pairs is governed by key overlap [10, 11]. Model editing operationalizes this on transformers: ROME inserts a fact by exactly the minimal rank-one update satisfying $W'k_* = v_*$ (up to key-covariance whitening) [12], and weight-space task vectors compose approximately additively [13].

L5 (fine-tuning is approximately kernel regression).
For small parameter updates, the function change at input x from training on x' is $\Delta f(x) \approx K(x, x') \times$(residual at x'), where K is the empirical neural tangent kernel $K(x, x') = \langle \nabla_\theta f(x), \nabla_\theta f(x')\rangle$ [14]; kernel dynamics have been shown to approximate language-model fine-tuning in several regimes [15].

3 The general rule

Suppose context c' originally produces behavior b' ($v_{b'} = f(c')$) and we fine-tune the model to instead produce b'' under c'. Two premises combine:

- **Write premise (L2, L3).** The fine-tune contributes a *near-constant write*: the same output-space direction $(v_{b''} - v_{b'})$ everywhere, with only its magnitude varying by context.
- **Gate premise (L5).** The fine-tune is a *small* update, so the per-context magnitude is first-order in the parameter change — a tangent-kernel similarity to the trained context.

Together these give the general rule: the induced change at any *bystander* context c is

$$\Delta v_b(c) \;=\; \underbrace{(v_{b''} - v_{b'})}_{\text{write: } \textit{what} \text{ leaks}} \;\cdot\; \underbrace{g(c)}_{\text{gate: } \textit{who} \text{ leaks}} \quad , \qquad g(c) \;=\; \frac{K(c, c')}{K(c', c')}, \tag{1}$$

with K the empirical NTK and $g(c') = 1$ by construction (the trained context gets the full write). Note the division of labor: the kernel view alone does *not* force a single output direction (for vector-valued f the NTK is matrix-valued); the constant-write factorization is the empirical content of L3, and the kernel supplies the scalar gate.

Multiple pairs. Training on several (context, behavior) pairs fits a kernel system: $\Delta f(c) = \sum_i K(c, c_i)\,\alpha_i$ with α solving the Gram equations on the training set. Contrastive negatives are the off-diagonal entries: persona contexts are mutually similar (large cross-kernel terms), so a positive-only implant writes nearly uniformly to all of them; negative pairs enter with opposing residuals and subtract this common mode, exposing the graded similarity term as a measurable leakage gradient.

The measured quantity. Measured leakage is a linear read of the write feature: at the read-out, $\Delta z_{\text{marker}} = \langle W_U[\text{marker}], \Delta v_b\rangle = \langle W_U[\text{marker}], v_{b''} - v_{b'}\rangle\, g(c)$, so log-probabilities inherit equation (1) up to softmax compression — one scalar slope per behavior, times the gate. Equivalently, emission behaves as a GLM: the known softmax nonlinearity composed with a model that is linear inside, which absorbs saturation and prior effects without changing the core.

What is actually assumed. The pre-trained map f cancels in trained$-$base differences; equation (1) constrains only the *edit*. The load-bearing assumptions are (i) the edit is small (lazy regime), and (ii) the edit's output direction is approximately constant across contexts (L3). Neither requires the model's own context-to-behavior computation to be linear.

4 The linear rung: the rank-one special case

The gate $g(c)$ is not directly observable without gradients, so we work down a ladder of proxies. The bottom rung linearizes everything: take the model's context-to-behavior wiring at one layer to be a linear map $v_b \approx Mv_c$ (best read as the local Jacobian $\partial v_b/\partial v_c$ of the frozen model). Then $\nabla_M f(c) = v_c$, the NTK reduces to the activation inner product $K(c, c') = \langle v_c, v_{c'} \rangle$, and equation (1) becomes

$$\Delta v_b(c) \; = \; (v_{b''} - v_{b'}) \cdot \frac{\langle v_{c'}, v_c \rangle}{\|v_{c'}\|^2}, \qquad \text{i.e.} \qquad M' \; = \; M \; + \; \frac{(v_{b''} - v_{b'})\, v_{c'}^\top}{\|v_{c'}\|^2}, \tag{2}$$

a rank-one read/write edit: read the trained context's key, write the behavior delta. Three independent routes give this same bottom-rung shape: (i) the minimal-norm edit fitting the new pair, $\min \|\Delta M\|_F$ s.t. $M' v_{c'} = v_{b''}$, which is ROME's update with whitened keys [12]; (ii) one delta-rule gradient step at the exact-fit step size, the classical outer-product update of associative memories [10, 11]; (iii) the linear-kernel case of equation (1). LoRA supplies the read/write structure by construction at the adapted layer ($\Delta W h = BAh$ is linear in the activation, A reads, B writes [2]).

The ladder between the rungs. Each factor of equation (1) can be relaxed independently:

(a) Gate proxies: linear read → distributional read → tangent kernel.
Cosine on mean activations is the linear kernel; Gaussian-KL adds second moments; MMD is a full kernel mean-embedding distance; the LoRA-restricted empirical NTK (one backward pass of the behavior logit w.r.t. adapter parameters per context) is the gate itself. The predictor bake-off has been climbing this ladder implicitly, and the gains over the linear read have so far been marginal (0.79 vs 0.76 on the marker; 0.50 vs 0.49 on sycophancy), suggesting the linear read captures most of the available signal once the layer is right.

(b) Write relaxation: rank one → rank r.
$\Delta v_b(c) = U\,\mathrm{diag}(s)\,V^\top v_c$. "One shared direction" weakens to "shifts live in an r-dimensional subspace"; the SVD we already run measures the effective rank directly.

(c) Gate relaxation: dot product → learned nonlinear gate.
The published gating mechanism is pre-existing QK attention reads [8] — bilinear plus softmax, so the true $g(c)$ is nonlinear in v_c. The constant write survives; predictability weakens from "computable from base cosine" to "estimable from a small sample of contexts," which is the per-behavior tuning the sycophancy result already suggested.

5 Conditional behavior without a conditional edit

The backdoor experiment of [8] pins down how a *conditional* behavior can be carried by a *constant* linear edit, and it is worth stating precisely because it refines the gate's mechanism. They train a single steering vector, added unconditionally at a fixed position (the last token before the response), on a Betley-style risky-backdoor task: give risky advice only when a jailbreak phrase appears in context. The resulting vector implements the conditional behavior on held-out in-distribution examples (validation accuracy ≈ 1.0) even though the write itself knows nothing about the trigger. Their proposed mechanism: value vectors on the trigger tokens *already* align with the risk direction in the base model; the learned vector shifts the last token's *queries* so its attention matches the keys at those trigger positions. Patching the last-token query vectors back to their base-model values kills the jailbreak, causally implicating QK attention. (Scope caveat: on this task both LoRA and the steering vector fail the out-of-distribution OOCR test, so the result establishes in-distribution conditionality, not generalized conditionality.)

Two refinements for our framework follow. **(i) The gate is attention-implemented.** The fine-tune's write can be a *query shift* that recruits a pre-existing read circuit, with the behavior content fetched from the context's own value vectors. After composing with attention, the net effect at the behavior slot still factorizes as write $\times$ gate — but the gate is a softmax attention pattern: potentially near-binary in the presence of a discrete trigger, and only smoothly similarity-like for diffuse contexts such as personas. This is the mechanistic case for rung (c) of the ladder. **(ii) The learned write need not match the prompted behavior direction.** The learned vectors have low cosine both with the "naïve" concept-difference vector (prompting the concept and subtracting) and with each other across seeds — many different vectors couple into the same pre-existing circuitry. The behavior's content lives in the base model; the edit only routes to it. This matches our own observation that the marker arm's shared shift direction is *not* the direction obtained by prompting the base model for the marker (#521), and it connects to the finding that the base model's own prior on the behavior carries predictive signal beyond context geometry (#532, #553): if leakage routes through base-resident representations, base-readable quantities about the *behavior in the context* are exactly what the geometry-only gate omits.

6 Relation to in-context learning

The rule plausibly covers contexts that arrive in the prompt, not just contexts trained on. Theoretically, a self-attention + MLP block provably converts its context into a *transient rank-1 weight patch* on the MLP [17] (extended to gated, bias-free, RMSNorm blocks of the Qwen/Gemma family by arXiv:2511.17864) — in this framework's terms, an ICL prefix performs the same write $\times$ gate operation as a fine-tune, applied for the duration of the context. Two consequences: the leakage rule should apply to prompt-elicited behavior shifts, and the context-side gate proxies should extend to ICL contexts. The second is already observed: base-model cosine/JS ranked marker leakage across ICL and multi-turn source contexts out of the box (#489).

The full "ICL = implicit fine-tuning" equivalence, however, is *not* available as a premise: for pretrained LLMs it is an explicitly open hypothesis. ICL and gradient descent differ in demonstration-order sensitivity and modify output distributions differently [19]; the similarity metrics behind the original equivalence evidence [18] are passed even by untrained models, and the two differ structurally in layer causality — the ICL "update" to a layer depends only on lower layers while GD's flows through the whole model [20]. So in-context behavior is a *proxy* for fine-tuning, with a validity question of its own, not a substitute.

What survives is a predictive, not mechanistic, link, and it is the most useful part. ICL on the same examples post-hoc recovers $> 85\%$ of the predictions fine-tuning corrects, in-distribution [18]; a model's own in-context inferences over a training corpus both anticipate fine-tuning's generalization failures (reversal curse under FT at near zero where ICL is near ceiling on the same data) and repair them when added to the training mix [21]; and for behavioral/stylistic adaptations the entire effect of in-context demonstrations compresses to a single steering-vector-like latent direction [22] — the same behavior-state object the write side of equation (1) uses. Forward prediction of fine-tuning *side effects* (leakage) from base-model in-context behavior is untested in the literature; that gap is P7.

7 What it implies

Each structural feature of equation (1) is a falsifiable prediction; rung-specific sharpenings are noted.

P0 (the adapter itself decomposes into read/write).
On the linear rung the trained update should be inspectable directly: per layer, the top right-singular vector of ΔW should align with the trained context's representation (read) and the top left-singular vector with the behavior delta (write) [2]. Testable on stored adapters with no new training. Caveat from §5: the write may align with a circuit-coupling direction rather than the prompted behavior delta (learned vectors in [8] had low cosine to the naïve concept-difference vector and across seeds) — a mismatch is informative about mechanism, not automatically a failure of the model.

P1 (shared write direction).
The activation shift at the behavior read-out points in the *same* direction $(v_{b''} - v_{b'})$ for every bystander context; only the scalar gate varies. This is the write premise (L3) tested in our regime, and it holds at every rung of the ladder. Operationally: stack per-bystander shift vectors, run an SVD, and the top component should dominate and match the pre-training behavior delta. Rank-r weakening: the shifts span a low-dimensional subspace.

P2 (the gate is a similarity to the trained context).
The per-bystander coefficient is $g(c) = K(c, c')/K(c', c')$. On the linear rung this is $\langle v_{c'}, v_c\rangle/\|v_{c'}\|^2$ — which explains why context cosine predicts leakage, and is *asymmetric*: the coefficient from $c' \to c$ is $\cos\theta \cdot \|v_c\|/\|v_{c'}\|$, not invariant under swapping source and bystander. Higher rungs replace the dot product with distributional reads or the LoRA-eNTK; the general claim is that *some* fixed similarity-to-source, computable from the base model, sets who leaks.

P3 (residual scaling; behavior dependence).
The update magnitude scales with $\|v_{b''} - v_{b'}\|$, the gap between the trained behavior and what the source context already produces. If the source context already elicits the behavior (e.g. a system prompt instructing it), the residual is near zero and the rule predicts little leakage even when context similarity is high; a similarity-only predictor gets this case wrong. On the read-out side, the softmax compresses $\Delta \log P$ near saturation, so prior-dependence should appear in log-probability space and shrink in logit space (the GLM view).

P4 (additivity / Gram structure).
Training on two pairs solves a two-point kernel system; when the cross-kernel term is small, leakages simply add (the activation-space analogue of task arithmetic [13]), and in general the deviation from additivity is predicted by $K(c_1', c_2')$.

P5 (layer locality).
The gate proxy should be read where the behavior is computed: late layers for a next-token marker, earlier for higher-level behaviors such as sycophancy. On the general rung this is a statement about where the tangent-kernel signal concentrates, not an extra assumption.

P6 (cross-behavior transfer).

The same update moves a *different* behavior b at a fixed context in proportion to the projection of $(v_{b''} - v_{b'})$ onto b's read-out direction, extending the rule from context generalization $(C, B \to C', B)$ to behavior generalization $(C, B \to C, B')$.

P7 (ICL proxy).

To the extent the transient-rank-1 picture of §6 holds, the base model's *in-context* behavior on the training mix should forward-predict the post-training leakage map: show the training examples in-context, measure behavior expression per bystander context, correlate with measured post-training leakage. This is the in-context analogue of the base-prior predictor that already beats geometry on instruction contexts (#532), and forward prediction of fine-tuning side effects from base ICL is unclaimed in the literature.

8 Regime of validity: lazy vs rich

Equation (1) and the entire proxy ladder are fixed-kernel, first-order pictures: they assume training adds a small edit while the model's features stay put (the lazy regime). If training moves the kernel itself (the rich, feature-learning regime), no base-geometry predictor works even in principle — the similarity that determines leakage is computed by features that no longer exist in the base model. This is the model's hard scope boundary, and it is diagnosable: measure NTK/feature drift between base and trained models, or watch base-geometry predictions degrade with training dose. It also reframes two questions as one: "how far is leakage predictable from base geometry?" and "when does fine-tuning leave the lazy regime?" The historical failures of geometric predictors on emergent misalignment have the signature of the rich regime.

9 How we are testing it

	Test	Status / result
P0	SVD of stored marker/EM adapters; compare top singular directions to $v_{c'}$ and to the measured behavior delta. External replication target: the per-trigger backdoor steering vectors in [8]'s repo, trained with the same behavior under 3–4 triggers but never cross-compared	Not yet run; free on artifacts we already have.
P1	SVD of per-context activation shifts after a marker implant vs misalignment SFT (#521, controls #551)	Surprising split: misalignment SFT shifts all 14 probed contexts along one shared direction (cos 0.96–0.98, $\sim$half the spectrum; seed-stable) — consistent with the convergent EM direction of [7]; the marker implant does not, and the trained context is the *least* aligned. Contrast survives controls. Benign SFT also yields one shared direction (#552, running), so the geometry may be generic to plain SFT; a positive-only marker retrain (#561, running) tests whether the contrastive rig, not the behavior, sets the geometry.
P2	Correlate base-model gate proxies with measured leakage across source$\times$bystander panels (#502, #489; sycophancy analogue). Kernel rung — LoRA-restricted eNTK gram vs cosine — not yet run	Context similarity ranks marker leakage well ($\|\rho\| \approx 0.76$–0.79 at layers 19–24) and extends to ICL/multi-turn contexts; distributional reads give marginal gains over the linear read; on held-out persona panels geometry predicts the training-induced *change* but the structure does not transfer across panels (#553), and the base-model prior beats geometric distance on instruction contexts (#532). The magnitude half of P2 is unsettled in #521.
P3	Vary how strongly the source context already elicits the behavior; instructed-marker break case; track logit vs log-prob spaces (#500, #531, #532)	Growing support: the bystander's own prior on the behavior is the most consistently supported predictor across panels; logit-space tracking is now in place to separate read-out compression from the update itself.
P4	Joint two-source training vs sum of singletons (#520, #527, #538)	Not yet diagnostic: additivity cosine reads 0.99, but each singleton's shift matrix is itself near rank-one, so near-parallel constants sum trivially; the confound gates fail at every tested training depth. A regime where singleton shifts vary across contexts is needed.
P5	Layer sweep of gate-proxy quality per behavior	Consistent: marker leakage is best predicted at layers 19–24; sycophancy at layers 1–8 (best cell: MMD at layer 7).
P6	Behavior-generalization testbed ($B \to B'$ leakage matrix, #545); context testbed (#537); formula-as-predictor (#510)	Queued: #537 running, #545 and #510 not yet run.
P7	ICL-vs-FT equivalence persona-framed (#491): K demos in-context vs SFT on the same K, equivalence at output + representation level; plus ICL-conditioned base responses as a before-training leakage predictor	Queued (#491 proposed); the leakage-prediction extension to ICL source contexts already holds (#489). Literature predicts task-type dependence: ICL $>$ FT on relational inference, FT $\geq$ ICL on classification at scale [21].

10 Known gaps

- The constant-write premise (L3) is borrowed from regimes that do not include persona-gated LoRA on Qwen-2.5-7B; P1 is its direct test here, and the marker arm currently *fails* it while the EM arm passes.
- The linear rung additionally assumes $v_b \approx M v_c$; a direct check is to regress v_b on v_c across many context–behavior pairs before training and measure how much variance a single linear map explains. Nobody has run this.
- The gate has only been proxied (cosine, Gaussian-KL, MMD); the gate itself (LoRA-restricted eNTK) has not been computed, so we cannot yet say whether predictor failures are failures of the model or of the proxy.
- The rule is stated at one layer; training updates all layers. P5 is a heuristic patch, not a derivation.
- The P1 results carry a training-rig confound (contrastive marker-only loss vs plain SFT) until #561 lands.
- The read-out is nonlinear (softmax over the unembedding), so activation-space predictions must be compared against logits, with log-probabilities treated as a compressed view near saturation.
- The strongest constant-steering evidence ([8, 9]) comes from one research orbit; and the published bridge from LoRA-induced trait shifts to persona directions did not survive our adversarial verification (only the prompt-elicitation projection result in [5] stands).
- Whether training *strength* raises the effective rank or context-dependence of the edit is unmeasured in the literature (the only published rank-raising lever is training-data diversity [9]); #538 found the geometry unchanged at a 3× deeper implant.

References

[1] N. Elhage et al. A mathematical framework for transformer circuits. *Anthropic (transformer-circuits.pub)*, 2021.

[2] J. Ward et al. Rank-1 LoRAs encode interpretable reasoning signals. arXiv:2511.06739, 2025.

[3] A. Turner et al. Activation addition: steering language models without optimization. arXiv:2308.10248, 2023.

[4] N. Panickssery et al. Steering Llama 2 via contrastive activation addition. arXiv:2312.06681, 2023.

[5] R. Chen, A. Arditi, H. Sleight, O. Evans, J. Lindsey. Persona vectors: monitoring and controlling character traits in language models. arXiv:2507.21509, 2025.

[6] M. Wang et al. Persona features control emergent misalignment. arXiv:2506.19823, 2025.

[7] A. Soligo, E. Turner, S. Rajamanoharan, N. Nanda. Convergent linear representations of emergent misalignment. arXiv:2506.11618, 2025.

[8] J. Engels et al. Simple mechanistic explanations for out-of-context reasoning. arXiv:2507.08218, 2025.

[9] J. Minder et al. Narrow finetuning leaves clearly readable traces. arXiv:2510.13900, 2025.

[10] T. Kohonen. Correlation matrix memories. *IEEE Trans. Computers*, 1972.

[11] J. A. Anderson. A simple neural network generating an interactive memory. *Mathematical Biosciences*, 1972.

[12] K. Meng, D. Bau, A. Andonian, Y. Belinkov. Locating and editing factual associations in GPT. *NeurIPS*, 2022.

[13] G. Ilharco et al. Editing models with task arithmetic. *ICLR*, 2023.

[14] A. Jacot, F. Gabriel, C. Hongler. Neural tangent kernel: convergence and generalization in neural networks. *NeurIPS*, 2018.

[15] S. Malladi, A. Wettig, D. Yu, D. Chen, S. Arora. A kernel-based view of language model fine-tuning. *ICML*, 2023.

[16] E. Hu et al. LoRA: low-rank adaptation of large language models. *ICLR*, 2022.

[17] B. Dherin et al. Learning without training: the implicit dynamics of in-context learning. arXiv:2507.16003, 2025.

[18] D. Dai et al. Why can GPT learn in-context? Language models implicitly perform gradient descent as meta-optimizers. arXiv:2212.10559, 2022.

[19] L. Shen, A. Mishra, D. Khashabi. Do pretrained transformers really learn in-context by gradient descent? arXiv:2310.08540, 2023 (ICML 2024).

[20] G. Deutch et al. In-context learning and gradient descent revisited. arXiv:2311.07772, 2023 (NAACL 2024).

[21] A. Lampinen et al. On the generalization of language models from in-context learning and finetuning: a controlled study. arXiv:2505.00661, 2025.

[22] S. Liu et al. In-context vectors: making in-context learning more effective and controllable through latent space steering. arXiv:2311.06668, 2023 (ICML 2024).